

Global View - 物理学 (C0282)

Global View - Physics (C0282)

对话 / Dialogue

Prof. Brown Good morning, everybody. Welcome to Physics 101. My name is Ed Brown, and I will be your professor for this semester. Since today is our first class, I wanted to give you an overview of what this course will look like, how you will be graded, and what we will cover this semester.

大家早上好。欢迎来到物理学入门课。我叫埃德·布朗，本学期由我担任你们的教授。今天是我们的第一节课，所以我想给大家介绍一下这门课的概况、你们将如何被评分，以及本学期我们会讲哪些内容。

Matt Will we be focusing more on theoretical physics or experimental physics, Professor?
教授，我们会更侧重于理论物理还是实验物理呢？

Prof. Brown This is an introductory course, and my aim is to give you a broad overview of the field of physics. The term “physics” encompasses many different areas of research and study, and I hope this course will provide you with conceptual understanding of physics, which will prove useful whether or not you choose to further your study in this field.

这是一门入门课程，我的目标是给你们一个物理学领域的宏观概览。“物理学”这个术语涵盖了许多不同的研究和学习领域，我希望这门课能让你们对物理学有概念上的理解，无论你们今后是否选择在这一领域深入学习，这都会很有用。

Prof. Brown We will begin the course by looking at the fundamental concepts of physics, then by the middle of the semester we will begin exploring the more theoretical side of physics. It is essential that you first have a firm grasp of the fundamentals, so that you can better understand the theoretical concepts when we get to them.

我们会先从物理学的基础概念讲起，到学期中段再开始探讨物理学更偏理论的一面。首先牢固掌握基础知识至关重要，这样当我们讲到理论概念时你们才能更好地理解。

Matt Will we learn about black holes, wormholes, and string theory?
我们会学到黑洞、虫洞和弦理论吗？

Prof. Brown We will learn about the general theory of relativity, including black holes. We will also explore developing theories in quantum mechanics, such as string theory. We will discuss some hypothetical features of space-time, like wormholes.

我们会学习广义相对论，包括黑洞。我们也会探讨量子力学中正在发展的理论，比如弦理论。我们还会讨论时空的一些假想特征，比如虫洞。

Prof. Brown We will also explore some of the more influential developments in the fields of thermodynamics, electromagnetism, and nuclear physics, all of which have had significant impacts on modern life. Now, I am going to have the TAs pass out the syllabus for this class, so you can see how this course will be graded.

我们还会探讨热力学、电磁学和核物理领域中一些较有影响力的进展，这些都对现代生活产生了重大影响。现在，我要让助教们把这门课的教学大纲发下去，这样你们就能看到这门课是如何评分的了。

Matt Oh man, looks like this isn't gonna be the easy A I thought it'd be!

哎呀，看来这门课不会是我原以为的那种轻松拿A的课了！

重点词汇 / Key Vocabulary

encompass	动词	把（某物）作为一部分包含在内，涵盖
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conceptual	形容词	基于或涉及思想或概念的，概念上的
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grasp	名词	对某事物的理解、掌握
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black hole	名词	外太空中一片不可见的区域，黑洞
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quantum mechanics	名词	研究原子层面能量与物质的物理学分支，量子力学
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hypothetical	形容词	涉及或基于假设的想法或理论的，假设的
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thermodynamics	名词	研究热之作用的科学，热力学
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electromagnetism	名词	由电流产生的磁场，电磁
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nuclear physics	名词	研究原子核的物理学领域，核物理学
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补充词汇 / Supplementary Vocabulary

academia	名词	大学及其教学人员的各种事务，学术界
applicable	形容词	可以用于现实生活情境的，适用的
gravity	名词	两个物体之间的吸引力，重力
ivory tower	名词	指与世隔绝、脱离现实、与外界缺乏联系或实用性的（学术）状态，象牙塔
relevant	形容词	可应用于日常情境的（知识），相关的、切合实际的